

Detecting Attention and Two Different Mind Wandering States Using Blink Frequency and Emotion Analysis

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SUMMARY Mind wandering is a cognitive state where attention shifts away from the primary task, affecting learning, productivity, and cognitive performance. We present a real-time, non-intrusive computer vision system that classifies Concentration, Deliberate (Intentional), and Spontaneous mind-wandering from eye aspect ratio (EAR) based blink dynamics and facial emotion recognition (FER), with contrast limited adaptive histogram equalization (CLAHE) for preprocessing. Our contributions are: adaptive EAR-based blink detection FER stabilization via eye angle alignment, CLAHE, and exponential moving average (EMA) smoothing; and decision-level temporal fusion of blink rate with FER-derived valence. Together, these advances enable robust real-time classification of concentration, deliberate, and spontaneous mind wandering, improving resistance to occlusion, noise, and illumination while maintaining temporal coherence. Experimental results demonstrate the effectiveness of this approach, with Random Forest achieving the highest classification accuracy of 99%, followed by Decision Tree at 97%. These findings indicate significant potential for applications in adaptive learning, driver monitoring, and cognitive research.

key words: mind wandering, blink frequency, emotion analysis, deep learning, cognitive state detection

1. Introduction

Mind wandering is a well-documented cognitive phenomenon that can have a significant impact on task performance, learning efficiency, and attention regulation [1]–[3]. Traditional methods for detecting mind wandering rely on EEG signals, self-reports, and fixed-interval analysis, which often lack real-time applicability [4]–[6]. This study proposes a deep learning-based, computer vision approach that integrates facial expression analysis, eye tracking, and blink frequency measurement to detect mind wandering in real-time. The dataset consists of participant responses recorded dynamically, allowing for accurate classification of cognitive states.

As humans, it is common for us to become distracted by

unimportant work or information while attempting to focus on an important task. For example, we may be listening to a lecture, reading a book, or performing other task-related tasks, when we become aware that our task-related thoughts have shifted from necessary to unnecessary thoughts such as what to eat for lunch, interacting with social media, news, or other life-related anxieties and worries. Mind wandering (MW) refers to attentional deviations from the task at hand, during which focus drifts to internal thoughts unrelated to the ongoing activity [1]. In this paper, we further distinguish between two widely recognized subtypes of MW: Deliberate (intentional) and Spontaneous (unintentional) (Carriere et al., 2013 [7]; Seli et al., 2016 [8]). Prior research has shown that such attentional drifting occurs frequently [9]–[11], with the frequency of mind wandering varying according to the nature of the task.

For example, one massive study of 5000 people from 86 countries discovered that 46.9% of the time, mind wandering occurred using data collected from an iPhone app [4]. MW is important to our daily lives because it often takes up half of our waking hours. There are advantages to MW, for sure [12]. MW is a crucial indicator of our sense of self and has been linked to original thinking [13]. It has the effect of assisting us in making the future. MW increased human creativity above and beyond the benefits of reading ability or fluid intelligence, as well as the general ability to solve problems or puzzles [6]. While MW has beneficial aspects, some adverse effects of MW were identified. In fact, it was discovered that a person in MW is less happy, indicating that a bad mood could be caused by a wandering mind [14]. Additionally, it endangers the safety of transportation, leading to a significant number of collisions and fatalities [15]. When we think about it, whether MW is good or bad depends on when we think about it and what we think about [16]. Therefore, daydreaming is not just an accidental occurrence. The incidents have a negative impact on a person's ability to do the task they are attempting. When performing hazardous jobs that call for extremely high levels of focus, patience, and tenacity, the repercussions of this negative influence might range from insignificant to life-threatening studies based on meta-analysis show a link between mind wandering and declines in human performance, including D'Mello et al. (2013), Randall et al. (2014). It goes without saying that multitasking will lead to decreased performance, especially if attention shifts from task-related to task-unrelated thinking. Evidence suggests that when

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attempting to find knowledge that is pertinent to a task being performed, the rate of inaccuracy increases because of mind wandering [4]. This decline in performance may have serious repercussions. It may have a negative impact on mental work like reading [4], [17]. Mind wandering can also result in less precise information recall [14], [18] when being tested to see how well a piece of written information was remembered. Mind wandering's negative effects can be mitigated proactively [19] or reactively [20]. Information and Communication Technology (ICT) has sped up progress and brought about significant changes. But there is still much work to be done in the difficult field of understanding human emotion. To make this happen, it is necessary to precisely identify a user's emotional state. Emotion is complex and mutable throughout time; it is dependent on a person's circumstances. The detection of emotions has been the subject of numerous research, some of which focused on behavior, language, social media, and facial expressions. Faces in the dataset and faces in the live video can be compared in order to identify and assess the intensity of the emotions. This method will be ideal for calculating emotional intensity via face recognition. The fer2013 dataset [21] is used in this study to calculate and recognize faces, as well as to extract emotions from human faces captured in live videos. The _mini XCEPTION.102-0.66.hdf5 model is used for emotion analysis and classification. Convolutional Neural Network (CNN) is used for image processing, and recognition is used in Artificial Neural Networks to process the pixels. CNN is a well-known image processing model that is used in AI and deep learning technologies [22]. In addition, the Keras library constructed the CNN and processed the image as a pixel, which takes the image and divides it into pixels or squares. And Opencv for detection of faces, which also detects features.

As stated above, MW has both beneficial and adverse facets. Studies have shed light on that feature of MW and pursued criteria to classify MWs. One of the influential criteria is the intentionality of MW, and intentional MW is referred to deliberate MW and unintentional MW is to spontaneous MW, respectively (Carriere et al., 2013; Seli et al., 2016). Deliberate MW is advantageous in the sense that it allows us to deploy our residual cognitive resource to irrelevant thoughts when the main task is less demanding or boring, without affecting task performance [7], [8]. It also allows us to come up with an idea by connecting thoughts that seem to be irrelevant at the first glance. In contrast, spontaneous MW is uncontrollable, which suddenly pops out in our mind and prevents us from sustaining attention on the primary task by diffusing our thoughts. Although psychology and clinical studies have frequently distinguished between Deliberate and Spontaneous MW, engineering research has seldom addressed this distinction, with only limited work investigating mind wandering through EEG or behavioral measures (e.g., Mittner et al., 2014; Dhindsa et al., 2019; Baldwin et al., 2017). When assessing attentional states of students or laborers by their physiological or expressive responses, it is crucial to discriminate deliberate MW from

spontaneous MW. Otherwise, we may interrupt students intentionally wandering their thoughts to get an insight, by providing inappropriate feedback. This study aimed to detect cognitive states by combining blink frequency and positive or negative emotions, and to evaluate classification accuracy using machine learning algorithms.

2. Methodology

This section outlines the methodologies employed to enhance the accuracy and efficiency of the proposed system. By leveraging deep learning methods, computer vision, and statistical analysis, the primary focus is on blink frequency and emotion detection (both negative and positive) for cognitive state classification. The proposed approaches are designed to ensure reliability, scalability, and robustness in real-world applications.

2.1 Dataset Description

This study uses real-time facial tracking data, which includes eye aspect ratio (EAR), blink frequency, and facial expressions. These characteristics are crucial for examining cognitive and attentional states. OpenCV and MediaPipe are used to collect the data [23], [24], guaranteeing frame-wise annotations for precise model training and validation. A thorough basis for deep learning model training in state classification tasks is provided by the dataset.

2.2 Data Collection

Videos taken while performing cognitive exercises make up the dataset utilized to detect mind wandering states. In order to extract blink metrics and face landmarks, the videos were analyzed frame-by-frame. We used MediaPipe Face Mesh [20] for landmark detection and OpenCV [23] for video frame extraction.

2.3 Data Preprocessing

A number of preprocessing methods are used to improve the data quality and model performance. First, photos are converted into a single-channel format via greyscale conversion, which lowers computing complexity [24]. Noise reduction is then applied using Gaussian filtering, which helps eliminate unnecessary artifacts and improves the clarity of extracted features [24]. Normalization is also implemented by scaling pixel values, ensuring uniformity and enhancing the model's convergence during training. Additionally, data augmentation techniques, such as synthetic variations of eye positions, are utilized to make the model robust to different lighting conditions and occlusions. These preprocessing steps collectively enhance the reliability and accuracy of the dataset [24].

2.4 Facial Landmark Detection

For accurate eye tracking and blink detection, we employed

MediaPipe Face Mesh, which defines 468 facial landmarks. From these, six key landmarks were selected for each eye to ensure precise tracking: Left Eye (33, 160, 158, 133, 153, 144) and Right Eye (362, 385, 387, 263, 373, 380). For clarity, Fig. 6 illustrates these landmarks on a sample facial image.

2.5 Eye Blink Detection

Eye blinks were detected using the Eye Aspect Ratio (EAR) with a participant-specific, slowly adaptive baseline rather than a fixed cutoff, improving robustness to illumination changes, occlusion, and individual eye morphology. The EAR is calculated using facial landmarks detected by the Media Pipe library [20]. The formula for EAR is:

$$EAR = \frac{|p2 - p6| + |p3 - p5|}{2 \times |p1 - p4|}$$

Where $p1, p2, p3, p4, p5$ are key points around the eye.

Blink Frequency Estimation. Blink detection is performed using the EAR threshold. A blink is registered when EAR falls below a predefined threshold for a continuous number of frames.

2.6 Emotion Classification

For six facial emotions—happy, sad, angry, surprised, fearful, and neutral, we used the *mini_XCEPTION* model for real-time FER [20], with eye-angle alignment, CLAHE normalization, and EMA smoothing to enhance robustness. These emotions were mapped to valence (with neutral tracked separately) and combined with EAR based blink rate using a rule-based 5-s state-persistence scheme to generate cognitive-state labels: Concentration, Deliberate Mind Wandering, and Spontaneous Mind Wandering. The model was pretrained on the FER-2013 dataset [21], and its architecture incorporated Global Average Pooling (GAP) to reduce overfitting. This model offers an optimal balance between accuracy and computational efficiency, making it well-suited for applications involving real-time human emotion tracking alongside physiological signals such as blink frequency.

The *mini_XCEPTION* architecture is derived from the Xception network [25], which is based on the principle of depthwise separable convolutions. These operations significantly reduce the computational cost of standard convolutions by decomposing them into two separate layers: depthwise convolution and pointwise convolution. In depthwise convolution, spatial filtering is applied independently to each input channel and is mathematically expressed as:

$$\text{Output}_{dw}(i, j, k) = \sum_{m,n} \text{Input}(i+m, j+n, k) \cdot K_{dw}(m, n, k)$$

Here, (i, j) are spatial positions, k is the channel index, and K_{dw} represents the depthwise convolution kernel.

Next, pointwise convolution is performed using 1×1 kernels across all channels, which integrates the depthwise

outputs into new feature maps. This can be written as:

$$\text{Output}_{pw}(i, j, c) = \sum_k \text{Output}_{dw}(i, j, k) \cdot K_{pw}(1, 1, k, c)$$

Combining both operations, the output of the model is:

$$\text{Output} = \text{PointwiseConv}(\text{DepthwiseConv}(\text{Input}))$$

The model accepts a grayscale input image of dimension $64 \times 64 \times 1$. The network begins with an initial convolutional layer followed by batch normalization and a ReLU activation. This is followed by four residual modules, each composed of depthwise separable convolutions with skip connections. These residual connections help to mitigate the vanishing gradient problem during training and improve convergence. The final convolutional layers are followed by a Global Average Pooling (GAP) layer, which helps reduce the risk of overfitting by minimizing the number of trainable parameters. The output is passed through a SoftMax layer to produce a probability distribution over the six emotion classes.

The softmax function used for the final classification is defined as:

$$P(y = i | x) = \frac{e^{z_i}}{\sum_{j=1}^6 e^{z_j}}$$

Where y_i is the ground truth label (one-hot encoded) and $\hat{y}_i$ is the predicted probability for class i .

In this research, we employed the pretrained model file named *mini_XCEPTION.102-0.66.hdf5*, which was trained on the FER-2013 dataset [21]. FER-2013 is a widely used benchmark dataset for facial emotion recognition, consisting of 35,887 labeled facial images divided into training, validation, and test sets. The model achieves a validation accuracy of approximately 66%, and due to its compact architecture, it is ideal for real-time applications where low latency is critical.

3. Experiments

3.1 Dataset

The dataset comprised 20 volunteers across 35 sessions, each lasting approximately 20 minutes. Probe responses were collected with a custom application that time-stamped and synchronized the self-reported attentional-state labels with the facial video stream and extracted features. Although subjective, probe responses remain the field's standard ground truth for mind-wandering. To assess inter-participant variability, we report per-participant means for blink frequency, emotion distributions, and model performance (see Fig. 7). Data acquisition included EEG recorded with the GES 400 Series Clinical Net Station at 250 Hz and facial expressions captured via webcam at 30 fps, an EEG device made by the E.G.I. Company, was used to record the participants' brain activity while they completed the task. In addition to EEG

measurements, a video recording tool was used to record participant faces. Our research is based on the dataset, which includes self-reported cognitive states (see in Behavioral Experiment), blink frequency, and facial expression data.

3.2 Behavioral Experiment

Twenty graduate and undergraduate students (13 females) enrolled in Shimane University participated in the experiment. The mean age of the participants was 19.28 (18yr - 25yr) and the standard deviation was 1.97. They reported normal or corrected to normal vision. Behavioral experiments were conducted in an electromagnetically shielded chamber (W2,059mm, D2,059mm, H2,150mm). A 23-inch monitor (EV2316W, EIZO Corporation) was positioned 70 cm from participants, with a chin rest used to maintain a stable viewing distance. This setup ensured reliable eye-tracking and video capture; although head movement was mildly constrained, spontaneous facial expressions were still captured effectively. The monitor, a keyboard, a mouse, and a web-camera (Logitech C920) were connected with a laptop PC (Dell Latitude E5540, Dell Inc.), which regulated stimulus presentation, response collection, and camera recording. Those functions were implemented by Psychopy software (Peirce et al., 2019). The monitor brightness was set to a moderate level, and the environment was adequately illuminated, ensuring clear visibility without affecting blink rates. To induce more MW, we located participants in a dimly lit room, where brightness was set to 840 lumens and color temperature was 3000K. However, the experiment room was not dark, and some outside light slotted into the room through the window of the door ahead which a black barricade was placed. Accordingly, the lighting was not overly dim, and we ensured that the room was adequately illuminated to allow for clear visibility of the facial features without inducing excessive eye strain or blink rate changes.

The participants performed two sessions of the counting Stroop task (Bush et al., 2006), each consisting of 576 trials. The task required participants to count the number of word items (1 - 4 items) while ignoring their meaning (i.e., one, two, three, four). The items were presented in Japanese (「いち」: one, 「二」: two, 「さん」: three, 「よん」: four), which subtended a visual angle of 1.23° (「二」)– 2.05° (「いち」, 「さん」, 「よん」) on the horizontal axis and 1.23° (one item)– 5.32° (four items) on the vertical axis. The number of items were always incongruent with their meaning, which induced response conflict. This all-incongruent design was chosen intentionally to avoid congruency-sequency effect, where reaction time becomes slow in an incongruent trial preceded by the congruent trial compared with an incongruent trial preceded by the incongruent trial (see Egner et al., 2007). Thenceforth, participants needed to sustain their attention to avoid error responses evoked by word meaning. In each trial, a cross fixation was presented on the center of the monitor for 1000ms, which was followed by a Stroop stimulus. The Stroop stimulus remained on the screen until a participant pressed a key or 1500ms elapsed. One of

three time intervals (i.e., 500, 750 or 1000ms) was placed between trials. In each session, probe stimuli were presented 20 times during the task, following Compton et al. (2019). Each probe asked participants to select one of six descriptions that best matched their attentional state just prior to the probe: (1) I fully concentrated on the task; (2) I performed the task while thinking about strategies or retrieving ideas to improve performance; (3) I thought about unrelated things without affecting task performance; (4) I thought about unrelated things that impaired task performance; (5) I was mind-wandered; (6) Other (e.g., I focused on an external sound). States (3) and (4) were used to distinguish between controlled mind-wandering, in which participants could maintain task performance, and uncontrolled mind-wandering, in which attentional control was lost and performance degraded. During the instructions, participants were explicitly told that option (3) referred to a state where MW was under control, whereas option (4) indicated a state where MW disrupted task performance. We classified a participant's attentional state as "concentrate", when they selected option (1) in response to the probe. Their attentional state was classified as "deliberate mind wandering" when option (2) or (3) was selected, and as "spontaneous mind wandering" as for option (4) or (5). When option (6) was selected, it was classified as "others".

3.3 Experiment Setting

In our experiment, we designed a system capable of real-time blink frequency detection and emotion classification from facial video input. Using selected facial landmarks, the Eye Aspect Ratio (EAR) was calculated to identify blinks per minute. Simultaneously, six facial emotions were classified using the pre-trained mini_XCEPTION model. These emotion classes were further grouped into positive and negative emotional states. The extracted blink frequency and categorized emotional data were then used as input features for machine learning algorithms to evaluate prediction accuracy for user attention states. The overall process is summarized in the following flowchart (Fig. 1).

3.4 Features Extraction Applications

Deep post-hoc analysis is made possible by the feature extraction application created in this work, which is intended for both structured data logging and real-time cognitive state monitoring. Over a predetermined period of time (5 seconds), it analyses facial video input and extracts important behavioral and emotional characteristics. The system uses MediaPipe Face Mesh to identify blinks and determine blink frequency in blinks per minute by detecting 468 face landmarks, with an emphasis on periocular areas. Simultaneously, a deep learning-based emotion identification model that has been trained on annotated facial emotion datasets uses facial region analysis to determine the probabilities associated with both positive and negative emotional cues as well as to identify the emotional state (neutral, happy,

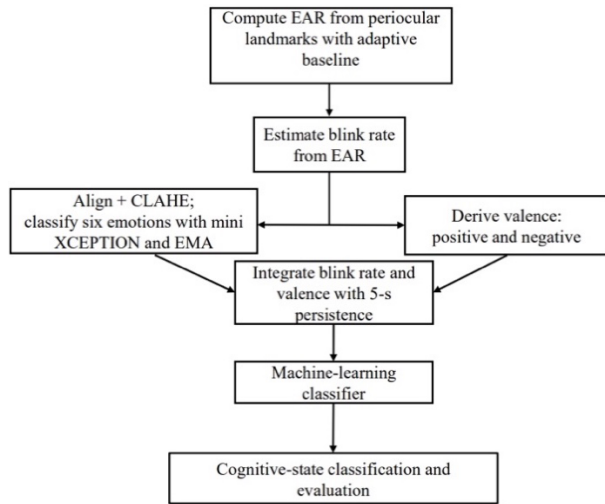


Fig. 1 Experimental framework



Fig. 2 Appearance of participants



Fig. 3 Experiment environment

sad, etc.). The extracted features—including blink count, blink frequency, emotion, emotional probability, and elapsed time—are then integrated to infer the user's cognitive state, which is categorized as Concentration, Deliberate (Intentional) Mind Wandering, or Spontaneous Mind Wandering.

3.5 Data Mapping

Before implementing machine learning models, data map-



Fig. 4 The GES 400 series clinical net station, E.G.I. Company.

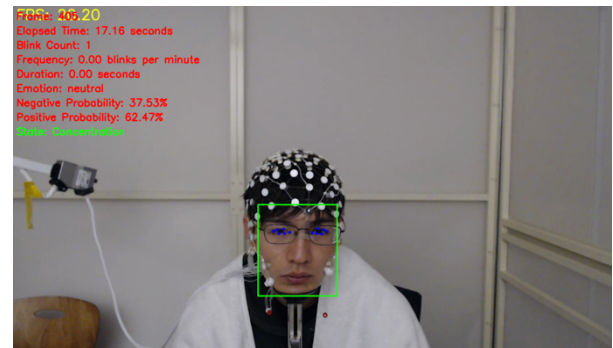


Fig. 5 Application interface

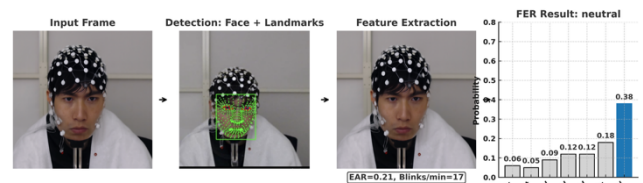


Fig. 6 Example image-processing pipeline and FER output

ping was used to correlate target labels (participants' reported cognitive states) with feature data such as blink frequency and facial emotion recognition. Each data sample included target data indicating focus, spontaneous mind wandering, or planned mind wandering, as well as feature data containing blink frequency, six emotion types, and positive or negative emotion states. This mapping ensured that the machine learning models correctly associated input features with target labels. The dataset was divided into training and testing subsets to determine the models' generalizability.

3.6 Results

Correlation Analysis

An initial correlation analysis (Fig. 8) between blink frequency and emotion (positive vs. negative) revealed no significant correlation. This suggested that, based on the correlation matrix, simple statistical relationships between blink frequency and emotion variables were insufficient for

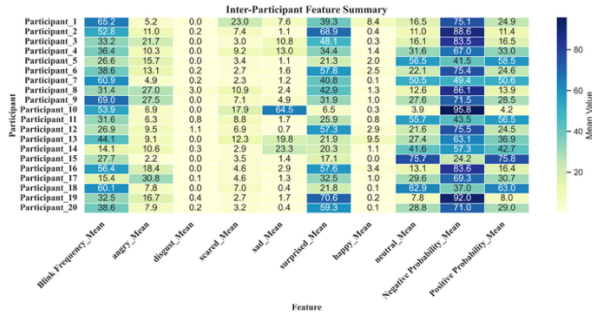


Fig. 7 Inter-participant variability of features.

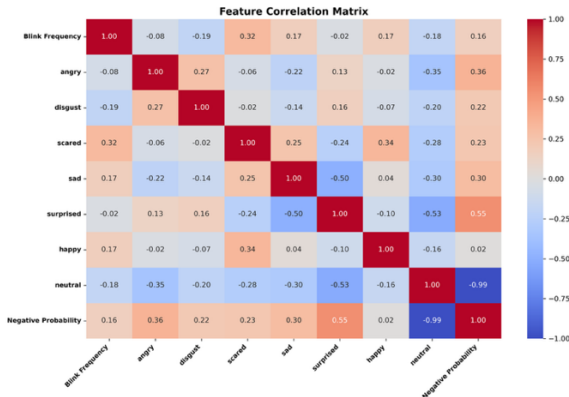


Fig. 8 Feature correlation matrix

predicting cognitive states. Future work should explore more advanced models, such as machine learning classifiers, to better capture these complex relationships. This finding underscores the need for more sophisticated modeling techniques to capture hidden patterns and dependencies in the data. Figure 7 shows inter-participant variability, with a heatmap of mean blink rate and EMA-smoothed emotion/valence probabilities for each participant.

Machine Learning Performance

We evaluated several machine learning algorithms, including Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Decision Trees, Random Forest, Gradient Boosting, Logistic Regression, and Naïve Bayes. Given the modest dataset size, classical ML models were selected for their robust generalization and interpretable feature contributions compared with end-to-end deep learning. For completeness, we report weighted and macro-averaged metrics, full classification reports, and confusion matrices which consistently confirm the superior performance of tree-based ensemble methods (see Table 1 and Figs. 9–10).

Among these models, Random Forest achieved the highest classification accuracy of 99%, followed by Decision Tree with 97%. The superior performance of Random Forest suggests that it effectively captures the non-linear relationships between features and cognitive states [24].

We analyzed behavioral data to see how attentional states affected performance of the task. When comparing

Table 1 Machine learning model performance

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.99	0.99	0.99	0.99
Gradient Boosting	0.79	0.79	0.79	0.79
Logistic Regression	0.49	0.49	0.49	0.49
Support Vector Machine (SVM)	0.76	0.76	0.76	0.76
K-Nearest Neighbors (KNN)	0.92	0.92	0.92	0.92
Decision Tree	0.97	0.97	0.97	0.97
Naïve Bayes	0.40	0.46	0.40	0.36

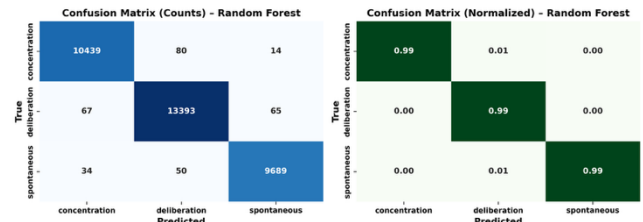


Fig. 9 Confusion matrix for random forest

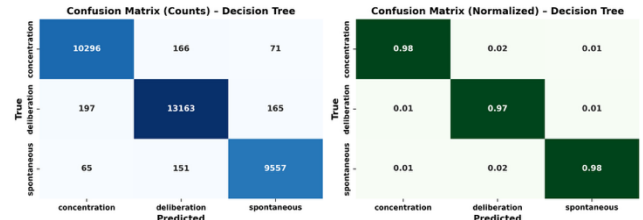


Fig. 10 Confusion matrix for decision tree

frequency of three attentional states (i.e., concentration, deliberate MW and spontaneous MW), they did not differ statistically. A repeated measure ANOVA did not show a significant main effect of the attentional state, $F(2, 38) = 0.95$, $p = .40$. The mean frequency of each attentional state was as follows: $M = 9.7$ times ($SD = 11.51$) for the concentration, $M = 14.75$ times ($SD = 7.25$) for the deliberate mind wandering, and $M = 12.8$ times ($SD = 9.65$) for the spontaneous mind wandering.

Because four participants reported no “concentration” state throughout the task, analyses on task performance (i.e., accuracy and response time) focused on two mind wandering states. For the analysis, one participant was removed because she selected the option (1) for all probes. Accuracy under the state of spontaneous mind wandering was lower than that under the deliberate. A paired t-test showed a significant difference between the mind wandering states, $t(18) = -3.89$, $p = .001$. An effect size analysis showed a large effect ($Cohen's D = 0.89$). The result ensured that spontaneous MW deteriorate performance of the Stroop task.

Response time under the state of spontaneous mind wandering was similar to that under that of deliberate mind wandering. A paired t-test did not show a significant differ-

ence between two mind wandering states, $t(18) = -1.72$, $p = .011$. An effect size analysis showed a medium effect (*Cohen's D* = 0.40). We further examined the two probe choices underlying this distinction—(3) unrelated thoughts without performance disruption and (4) unrelated thoughts that impaired performance. Accuracy was significantly greater for choice (3) ($M = 0.96$, $SD = 0.06$) than for choice (4) ($M = 0.87$, $SD = 0.15$), $t(18) = 3.81$, $p = .001$, while RTs did not differ significantly between the two, $t(18) = -0.77$, $p = .45$. This suggests that participants reliably discriminated between the two options based on their subjective sense of controllability. Table 1 summarizes model performance: Random Forest achieved the highest accuracy and F1 score (0.99), followed by Decision Tree (0.97). Figures 9 and 10 present confusion matrices for Random Forest and Decision Tree, respectively, with left panels showing raw counts and right panels showing row-normalized proportions, highlighting residual confusions primarily between the two mind-wandering classes.

4. Discussion

The study highlights the effectiveness of machine learning in detecting attention and mind-wandering states using blink frequency and facial expressions. Initial correlation analysis found no significant direct relationship between blink frequency and emotion, emphasizing the need for machine learning to identify underlying patterns. Among the tested models, Random Forest achieved the highest accuracy (99%), followed by Decision Tree (97%). Both effectively captured complex, non-linear relationships between facial expressions, blink patterns, and cognitive states, with Random Forest's ensemble approach providing stronger generalization. Gradient Boosting (79%) and KNN (92%) showed moderate performance, whereas Logistic Regression (49%) and Naïve Bayes (40%) performed poorly, confirming that simple probabilistic and linear models fail to capture the nuanced dependencies in the data. Feature importance analysis revealed that blink frequency and emotions like 'neutral', 'sad', and 'surprised' were strong indicators of cognitive states. Additionally, negative emotions played a crucial role in distinguishing spontaneous and deliberate mind-wandering. Spontaneous mind-wandering exhibited more emotional variability, whereas deliberate mind-wandering showed more controlled expressions. In future work, to address the lack of external ground truth, we will expand the cohort and incorporate multimodal validation, synchronized EEG and expert-annotated blink/emotion events, to corroborate self-reports and evaluate more expressive deep learning architectures.

5. Conclusion and Future Work

Our findings align with theories of cognitive control and attention regulation, indicating that emotion and blink frequency are key indicators of cognitive states. They contribute to the broader framework of affective computing by

offering insights into how emotion regulation may shape mind-wandering and task performance. Further investigation of these theoretical links is needed to refine the model and extend its applicability. This study introduced a non-intrusive approach for detecting mind-wandering states by analyzing blink frequency, eye movement, and emotion recognition using deep learning techniques. Unlike traditional fixed-interval methods, our approach utilizes real-time participant responses for a more dynamic and accurate classification of cognitive states into Concentration, Deliberate/Intentional Mind Wandering, and Spontaneous Mind Wandering. Among various machine learning models tested, Random Forest achieved the highest classification accuracy of 99%, followed by Decision Tree with 97%, highlighting the effectiveness of tree-based ensemble methods. The findings demonstrate the potential applications of this method in adaptive learning, driver safety monitoring, workplace productivity, and cognitive neuroscience research. However, to enhance the generalizability and robustness of the model, future work should explore deep learning techniques, larger datasets, and the integration of EEG data to further enhance classification accuracy and provide more objective validation of cognitive states.

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